MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY, CHARGE STORAGE

1/ TYPE 7423

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Direct-view charge storage tube, 5-inch, electrostatic focus and deflection, aluminized, P20 phosphor

DIMENSIONS AND PIN CONNECTIONS: See figures 1 and 3

ABSOLUTE RATINGS:

Viewing gun (see note 1)

Parameter:	Ef	ek	Eb1	Eb2	Eb3	Eb4	E(coll)	Ebe	ebe	E(scrn)
Unit:	V	vdc	Vdc	Vdc	Vdc	Vdc	Vdc	Vdc	v	kVdc
Maximum:	6.9	0	80	150	150	150	250	---	25	10 (Note 5)
Minimum:	5.7	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	0 (Note 4)	60	Adjust (Note 2)	Adjust (Note 2)	90	180	10	Adjust (Note 3)	8.5

ABSOLUTE RATINGS:

Writing gun (see note 6)

Parameter:	Ef	Ehk	Ek	Ec1	Ec2	ed	Ec3
Unit:	V	Vdc	Vdc	Vdc	Vdc	vdc	Vdc (Note 15)
Maximum:	6.9	+125	-1,000	0	150	500	500
Minimum:	5.7	-125	---	-150	---	---	0
TEST CONDITIONS:	6.3	0	-540	Adjust	0	---	Adjust

GENERAL:

Qualification - Required
Stabilization - See note 7
1/ See note 26

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
Ⓒ 1216	Base material insulating quality	-		---	---	---	---
1141	Pressure (implosion)	-		---	---	---	---
1002	Barometric pressure, reduced	21		---	---	---	---
1026	Low-temperature operation	-		---	-55	---	°C
1026	High-temperature operation	-		---	---	71	°C
---	Vibration	22		---	---	---	---
---	Shock	23		---	---	---	---
	Quality conformance inspection, part 1						
5401	Voltage breakdown	-		---	---	---	---
5401	Voltage breakdown (multigun types)	8		---	---	---	---
---	Anode No. 1 current	2,9		Ib1	0.5	5.0	mAdc
---	Anode No. 2 voltage	2		Eb2	25	40	Vdc
---	Anode No. 2 current	2,9		Ib2	---	1.5	mAdc
---	Anode No. 3 voltage	2		Eb3	10	25	Vdc
---	Anode No. 3 current	2,9		Ib3	---	1.5	mAdc
---	Anode No. 4 current	2,9		Ib4	---	1.5	mAdc
---	Collector current	2,9		I(coll)	---	1.0	mAdc
---	Screen current	2		I(scrn)	---	2.0	mAdc
---	Cathode current (view gun)	-		Ik	2.5	10.0	mAdc
Ⓒ 5412	Luminance (light output), visual output	10	Method A	---	4,000	---	ftL
Ⓒ 5409	Blemishes, visual output	11		---	---	---	---
---	Display cutoff variation	12		---	---	20	%
---	Storage time	13		---	20	---	Sec
Ⓒ 5416	Erasing time, visual output	14		---	---	20	msec
---	Writing speed	24		---	27,000	---	In./sec
---	Pattern distortion	25		---	---	0.050	Inch
5246	Focus voltage at cutoff	-		Ec3	75	150	Vdc
5246	Focus voltage at zeros bias	-		Ec3	75	150	Vdc
Ⓒ 5402	Grid-cutoff voltage	16		Eco	30	60	Vdc
Ⓒ 5404	Spot position (electro-static deflection)	17	Method B	---	---	---	---
5231	Spot displacement (leakage)	-		---	---	10	mm
5248	Deflection factor	-	1D2	DF	29	35	Vdc/in.

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1 -Continued						
5248	Deflection factor	-	3D4	DF	28	34	Vdc/in.
5248	Deflection-factor uniformity	-		---	---	---	---
5101	Base alinement (electrostatic types)	18	+1D2	---	---	---	---
5101	Side terminal alinement (electrostatic types)	-	See figure 1	---	---	---	---
5101	Side terminal and base alinement	-	See figure 1	---	---	---	---
5101	Angle between traces	-		---	88	92	Degrees
Ⓒ 5411	Resolution, visual output	19		---	25	---	Lines/in.
5251	Heater-cathode leakage current	-		---	---	50	µA
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Grid No. 2 leakage current	-		---	---	---	---
---	Viewing time	20		---	2	---	sec
	Quality conformance inspection, part 2						
1301	Heater current (view gun)	-		If	1,260	1,540	mA
1301	Heater current (write gun)	-		If	540	660	mA
5216	Stray light emission (conventional types)	2		---	---	---	---
1331	Direct-interelectrode capacitance	-	Writing gun				
			Cathode to all	Ck	---	7.0	pF
			Grid No. 1 to all	Cg1	---	11.0	pF
			D1 to D2	C1D2	---	5.0	pF
			D3 to D4	C3D4	---	3.5	pF
			D1 to all	CD1	---	12.0	pF
			D2 to all	CD2	---	11.0	pF
			D3 to all	CD3	---	8.0	pF
			D4 to all	CD4	---	9.0	pF
			be to all	Cbe	---	50.0	pF
Ⓒ 1101	Secureness of base, cap, or insert	-		---	---	---	---
Ⓒ 1111	Base pin solder depth	-		---	---	---	---
Ⓒ 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group D; light output = 1,000 ftL written and erased at a rate of 60 Hz; t = 500 hours (min)	---	---	----	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3 -Continued						
---	Life-test end points:						
5412	Luminance (light output), visual output	12		---	3,200	---	ftL
---	Storage time	13		---	10	---	Sec

NOTES:

1. Viewing gun voltages are referenced to the viewing gun cathode which is generally operated at ground potential. However, there are some modes of operation where it is desirable to pulse off the viewing beam by applying a positive pulse to the cathode.
2. Collimate by adjusting for maximum display size and uniformity. These conditions can be closely approached when, as the display size is increased, the phosphor current reaches a maximum.
3. Adjust pulse height and duty cycle for test - It varies with the test.
4. In one mode of operation the view gun cathode is pulsed to collector potential to obtain increased viewing time.
5. A series current-limiting resistor of 1 megohm shall be used in the viewing screen high-voltage lead. E(scrn) is measured on the tube side of the 1-megohm resistor.
6. Writing gun voltages are referenced to the writing gun cathode except Ek and Ec2 which are referenced to the viewing gun cathode.
7. All tubes shall be stabilized for a period of 22 hours with both heaters on and the viewing system fully operating. A 6.2 volt ac pulse is applied to the backing electrode during this period.
8. The electrodes of the writing gun to be set at maximum rated voltages are Ec2 and Ec3, since Ec2 corresponds to a cathode ray tube's anode No. 2.
9. Electrode currents vary with the voltage on the insulator surface; the maximum and minimum values specified herein are under the appropriate extreme conditions.
10. The brightness of the viewing screen measured when saturation current is flowing to the viewing screen and viewing screen voltage is 8.5 kV ±5 percent shall be not less than the value specified herein.
11. The faceplate shall be of a normal optical standard and shall be macroscopically free of scratches, stones, or striae. Under the following conditions, examination of the faceplate in the area indicated on figure 4 shall be made. Operating conditions to obtain a minimum viewing time of 2 seconds. The erasing conditions for this viewing time are obtained with a train of 1.5 µs pulses, with amplitude adjustable for 0 to 10 volts, with the prf variable from 500 to 1,500 pps. Blemishes shall not exceed the limits specified below:

Dark spots	Number
0.005 to 0.017 inch (0.13 to 0.43 mm) diameter	10 maximum
0.017 to 0.032 inch (0.43 to 0.81 mm) diameter	3 maximum
0.032 inch (0.81 mm) and over	0

Brilliant spots. Spots which are as bright or brighter than the maximum brightness and cannot be erased. None acceptable.

Bright spots. Spots which can be erased by an erase duty cycle 10 percent greater than required for viewing time.

0.017 inch (0.43 mm) diameter or less	2 maximum
0.017 to 0.032 inch (0.43 to 0.81 mm) diameter	1 maximum

12. The saturated brightness display is erased with a train of high duty cycle (>50 ms per second) erase pulses of an amplitude just sufficient to cause some area of the display to reach cutoff. This amplitude is termed L. The erase pulse amplitude shall be increased until all of the display is cut off. This amplitude is greater than L by an amount D. The variation is specified by $\frac{D}{L} \times 100$ percent.
13. Storage time is time required for the brightness to increase from cutoff to 50 percent of its saturation value. No erase pulses are applied during this time.

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NOTES: -Continued

14. The time required to bring the display from saturation brightness to cutoff by pulsing the insulator support screen. The maximum erase pulse amplitude shall be 10 volts.

15. This is the focus voltage range for best focus for 20-volt grid drive.

16. Grid voltage for visual extinction of the undeflected, stored spot for 5 seconds observation.

17. The spot shall fall within a circle of 0.2 inch (5.08 mm) radius centered 0.6 inch (15.24 mm) above the center of the tube face.

18. See figure 3. Spot is deflected along a line passing between pins 9 and 10 or 230 degrees clockwise from the key.

19. Resolution is measured at the 90 percent of saturation brightness by the shrinking raster method at the center of the tube.

20. Under the test conditions with erase pulses applied to counteraction charging of the storage surface, this is the time required for a signal written to saturation brightness, anywhere in the display to decay to cutoff. Erasure for viewing time measurement shall be accomplished by a train of 1.5 µs pulses, adjustable 0 to 10 volts amplitude, over the range 500 to 1,500 prf.

21. Continuous operation under the test conditions barometric pressure ranging from 30 inches (762.0 mm) of Hg to 2 inches (50.80 mm) of Hg (60,000 feet), the pressure remaining constant for 2 minutes or varying at a rate as high as 4 inches (101.60 mm) of mercury per second. There shall be no sign of arc over or harmful corona at the base pins or viewing screen.

22. The tube shall be held rigidly in an appropriate fixture. The tube shall be subjected to a sinusoidal vibration from 5 to 500 Hz at 3 G ±1 G (0.30 inch (7.62 mm) total excursion below 18 Hz.) Conditions and duration of the test shall be in accordance with the vibration test of MIL-T-5422. The tubes shall be operating during and after the test.

23. The tube shall be held rigidly in the appropriate fixture. (Conditions 1, 3, and "steady acceleration" are checked on Barry Shock Machine, Model 150VD, or equivalent. Condition 2 is checked on the Taft-Pierce High Impact Machine No. 2, or equivalent with the hammer modified with a felt pad to obtain the desired impact shocks.)

    Condition 1. 18 shocks of 15 G ±10 percent peak value consisting of three shocks in opposite directions along each of three mutually perpendicular axes, each shock impulse having a time duration of 40 ms ±50 percent. (Peak value at duration's midpoint ±10 percent.) The tube shall be operating during and after the test.

    Condition 2. 6,000 impact shocks, 1,000 shocks of 25 G ±10 percent peak value in opposite directions along each of three mutually perpendicular axes, each shock impulse having a time duration of 5 ms ±50 percent. The tube operating during and after the test.

    Condition 3. 12 impact shocks of 30 G ±10 percent peak value consisting of two shocks in opposite directions along each of three mutually perpendicular axes, each shock impulse having a time duration of 11 ms ±10 percent. There shall be no evidence of breakage of glass envelope as a result of this test.

    Steady acceleration, nonoperating, 8.05 G for 1 minute in each of six directions. The tube shall be operable after test.

24. With the display erased to cutoff, a raster of separate lines is written on the tube using 20 volts video drive from the bias level where the beam current is 0.5 µA. At the scanning speed specified, the trace brightness shall be equal to 1,500 ftL or 35 percent of saturation brightness (whichever is greater). The writing speed shall be measured in approximately the position shown on figure 5. The saturation brightness for this test shall be measured in the 1 square inch (6.45 sq cm) area shown on figure 5.

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NOTES: -Continued

25. Some trace curvature will be evident. This will be greatest on the side of the display farthest from the offset writing gun's axis. The trace curvature, pattern distortion, is measured at the location shown on figure 5. The pattern distortion value is measured outside of the 4-inch (101.6 mm) diameter quality circle, but still visible within the tube's 4.25-inch (107.95 mm) display area. The viewing system is operating, but the support screen is biased below cutoff.

26. The tube shall be operated in a suitable magnetic shield.

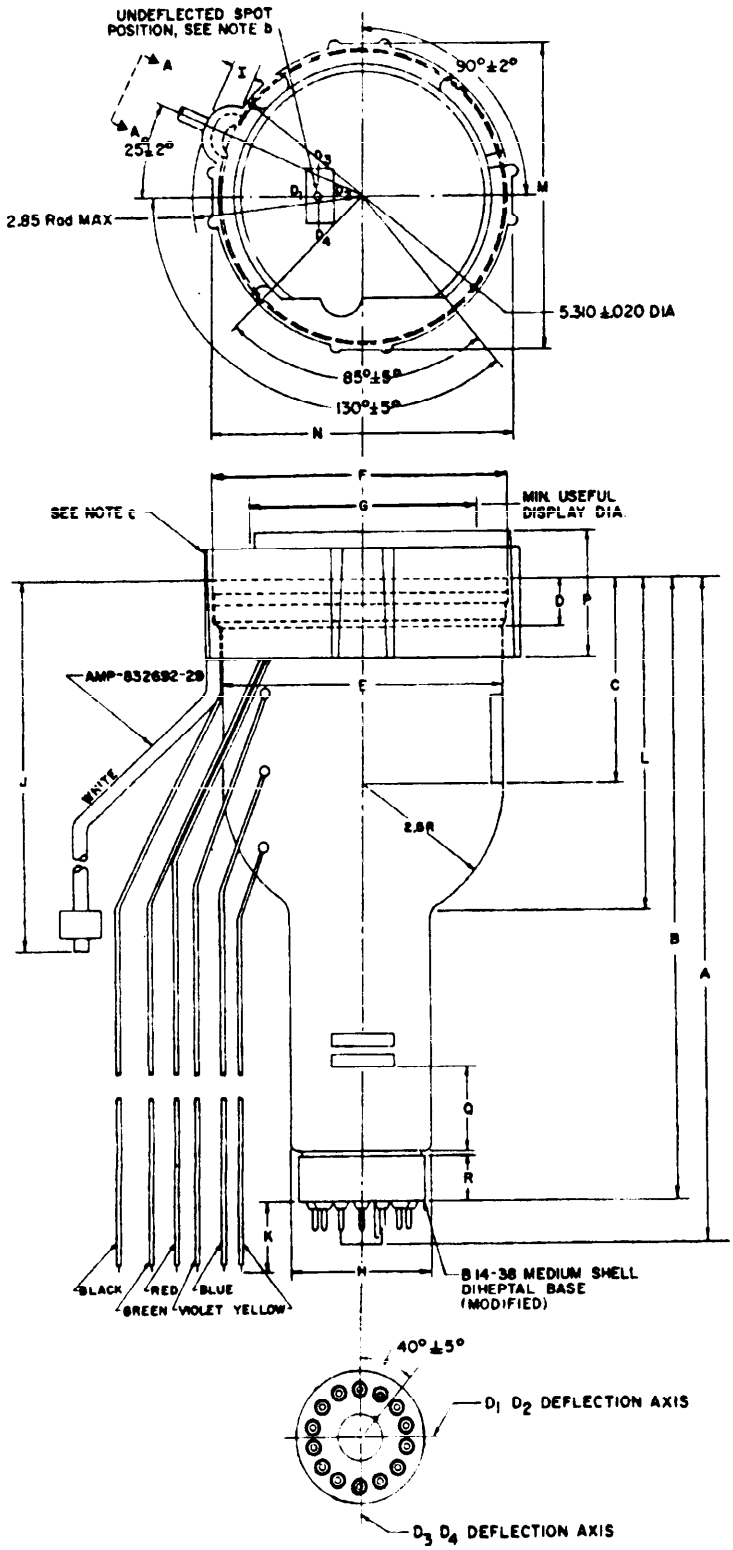
User activities:
Navy - AS, OS, MC, CG, SH

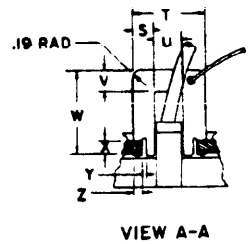
Preparing activity:
Navy - EC

Agent:
DSA - ES

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INCHES	MM
.020	.51
.19	4.83
2.50	63.50
2.85	72.39
5.310	134.87

FIGURE 1. Outline drawing of electron tube type 7423.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		12.062 (306.37)
B	11.125 (282.58)	11.250 (285.75)
C	3.730 (94.74)	3.930 (99.82)
D	.690 (17.53)	.810 (20.57)
E	4.940 (125.48)	5.060 (128.52)
F	5.297 (134.54)	5.328 (135.33)
G	4.000 (101.60)	
H	2.440 (61.98)	2.560 (65.02)
I	.355 (9.02)	.395 (10.03)
J		8.250 (209.55)
K	4.500 (114.30)	
L	5.840 (148.34)	5.970 (151.64)
M	5.320 (135.13)	5.360 (136.14)
N	5.500 (139.70)	5.540 (140.72)
P	2.260 (57.40)	2.300 (58.42)
Q	1.484 (37.69)	1.516 (38.51)
R	.797 (20.24)	.828 (21.03)
S	.375 (9.53)	.385 (9.78)
T	1.245 (31.62)	1.255 (31.88)
U	.495 (12.57)	.505 (12.83)
V	.375 (9.53)	.385 (9.78)
W	1.495 (37.97)	1.505 (38.23)
X	.295 (7.49)	.305 (7.75)
Y	.055 (1.40)	.065 (1.65)
Z	.155 (3.94)	.165 (4.19)

NOTES:

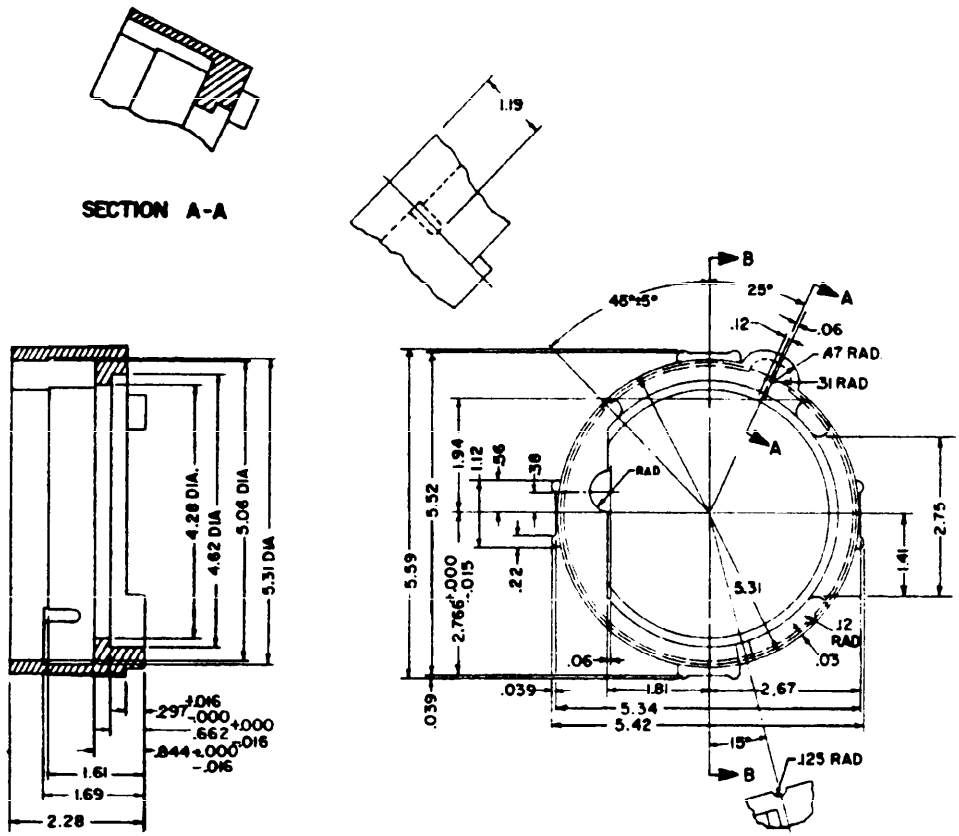
a. Deflecting electrodes D1, D2 are nearer the screen; deflecting electrodes D3, D4 are nearer the base with D1 positive with respect to D2, the spot will be deflected toward pins No. 9 and 10; likewise with D3 positive with respect to D4, the spot will be deflected toward pin No. 6.

b. The spot shall fall within a circle of 0.2 inch radius centered 0.6 inches above the center of the tube face.

c. Insulating ring shall be as shown on figure 2. The high voltage potting ring shall be covered with a colloidal graphite film. The ground strap detailed in view A-A shall be installed to make contact with the colloidal graphite film.

FIGURE 1. Outline drawing of electron tube type 7423 - Continued.

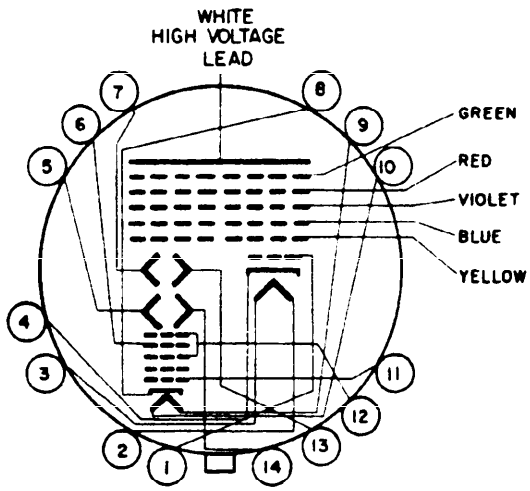
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INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.015	.38	.22	5.59	.844	21.44	1.94	49.28	5.06	128.52
.016	.41	.297	7.54	1.12	28.45	2.28	57.91	5.31	134.87
.03	.76	.31	7.87	1.19	30.23	2.67	67.82	5.34	135.64
.039	.99	.38	9.65	1.41	35.81	2.75	69.85	5.42	137.67
.06	1.52	.47	11.94	1.61	40.89	2.766	70.26	5.52	140.21
.12	3.05	.56	14.22	1.69	42.93	4.28	108.71	5.59	141.99
.125	3.18	.662	16.81	1.81	45.97	4.62	117.35		

FIGURE 2. Insulating ring.

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14 PIN SOCKET

PIN NO.	1	2	3	4	5	6	7	8	9	10	11
ELEMENT:	VA_1	VH	VH	VK	D_3	WG_3	D_1	WK	WH	WH	WG_1

PIN NO.	12	13	14
ELEMENT:	WG_2	D_2	D_4

LEAD	GREEN	YELLOW	BLUE	VIOLET	RED	WHITE HIGH VOLTAGE LEAD
ELEMENT:	Be	A_2	A_3	A_4	COLL.	VS

FIGURE 3. Basing diagram and pin connections.

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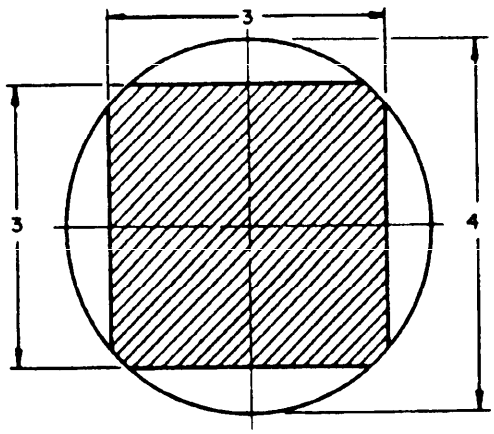


FIGURE 4. Faceplate quality area.

INCHES	MM
1.00	25.40
1.25	31.75
1.323	33.60
2.000	50.80
3.000	76.20
4.00	101.60

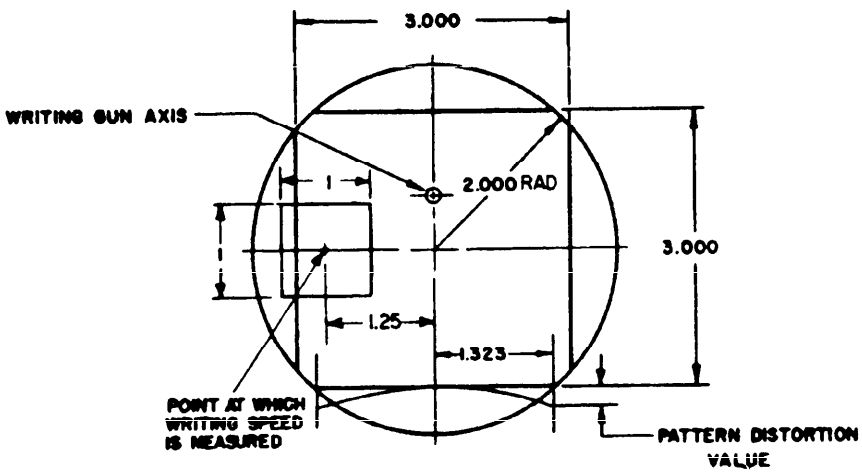


FIGURE 5. Pattern distortion.

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